Learning Patterns

Introduction to NLP — MSc. DH ENC-PSL

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1 Recap

1.0.1 The Language Modeling Problem

Language Model

A Language Model (LM) estimates the probability of pieces of text. Given a sequence of text $w_1, w_2, \cdots, w_S$, it answers the question:

What is $P(w_1, w_2, \cdots, w_S)$?

How to compute P?

- Count-based (count): n-gram
- Neural method (learn): Transformer
  - Masked Language Modeling (encoder-only, BERT)
  - Causal Language Modeling (decoder-only, GPT)

→ encode:

- gramaticality
- semantic plausability
- stylistic consistency
- knowledge (?)

1.0.2 Word Embeddings — Feature Extraction

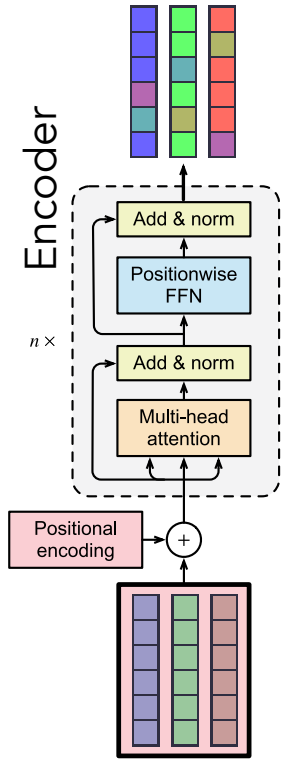


Diagram encoder-only model (e.g.; BERT).

- Tokenize text sequence
  - BERT means Bidirectional Encoder Representations from Transformers.
  - [CLS] bert means bid ##ire ##ction ##al en##code ##r representations from transformers . [SEP]
- Run through the pre-trained LM
  - Output embeddings: $(\mathrm{N_{batch}}\times)\mathrm{N_{tokens}}\times\mathrm{d_{model}}$
- Extract contextual embedding at the poisition of choice

1.0.3 Document Representation & Unsupervised Tasks

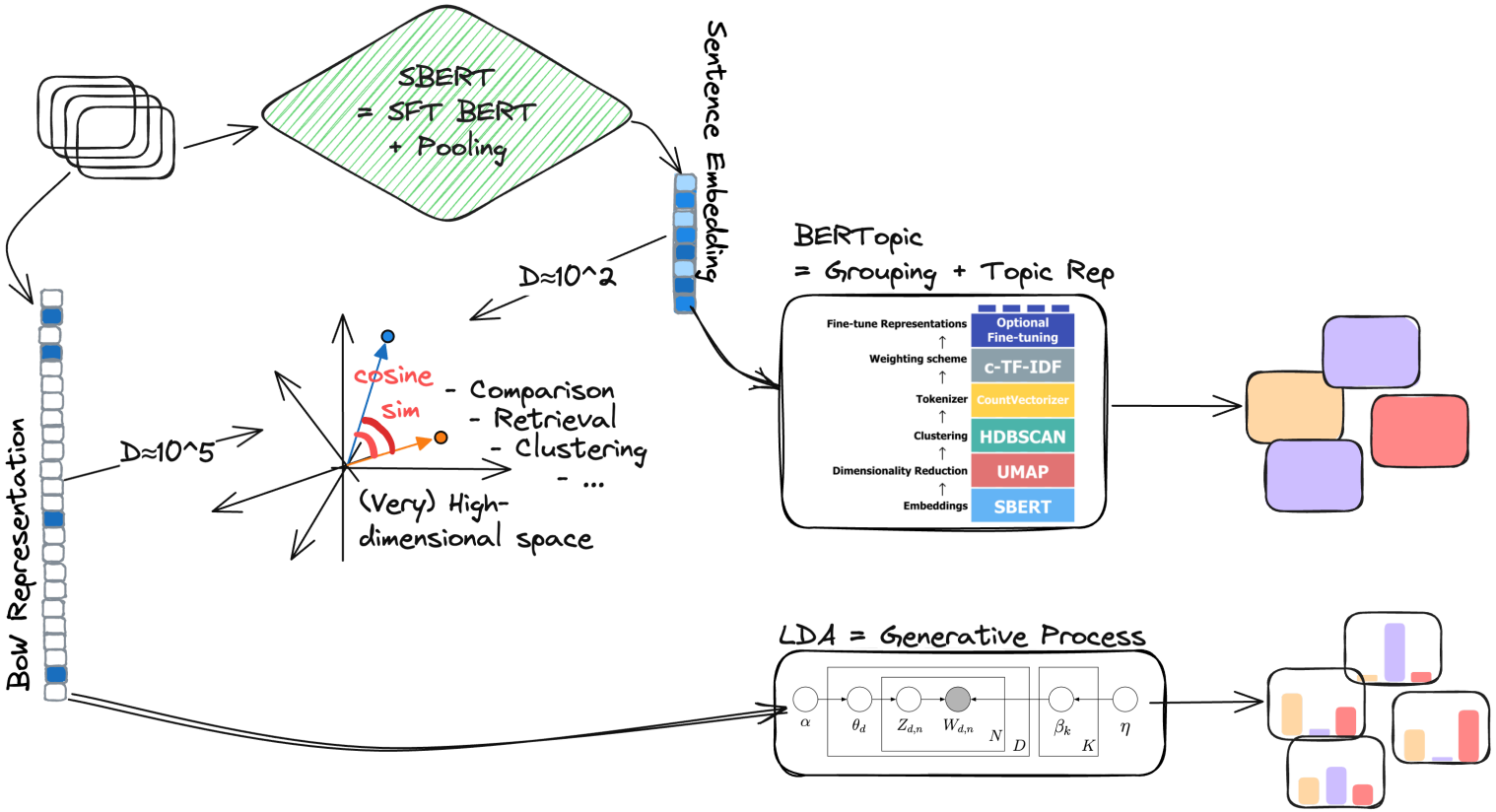


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2 Supervised Learning

2.1 Introduction

2.1.1 Supervised Learning



Definition

Learn patterns that map *inputs* (texts, sentences, etc.) to *outputs* (labels, categories, values) based on *annotated examples*.

What you need?

- **Training data**: documents annotated with *labels* ('ground-truth')
- A **model** that can make predictions
- A **loss function** that measures the model performance
- An **optimisation process** that nudge the model towards lower loss

→ The model will '*learn*' from examples (and corrections), improving its predictions through training.




2.1.2 Supervised Learning NLP Tasks

- **Classification**: Assign a *label* to an entire text or sentence
  - *Sentiment analysis* (positive / negative reviews)
  - *Topic classification* (political / cultural / literary texts)
  - *Spam detection*, *author identification*, *genre prediction*
- **Regression**: Predict a *continuous value*
  - Predicting a *readability score* or *publication year* from a text
- **Sequence Labelling**: Assign a label to *tokens* in a sequence (classification)
  - *Named Entity Recognition (NER)*: tagging names of people, places, organizations
  - *Part-of-Speech tagging*
- **Question Answering (QA)**
  - The model identifies an *answer span* in a text based on a given question
- **Other tasks**
  - *Text summarization*, *machine translation*, *paraphrase detection*
    (still supervised — but outputs are more complex)



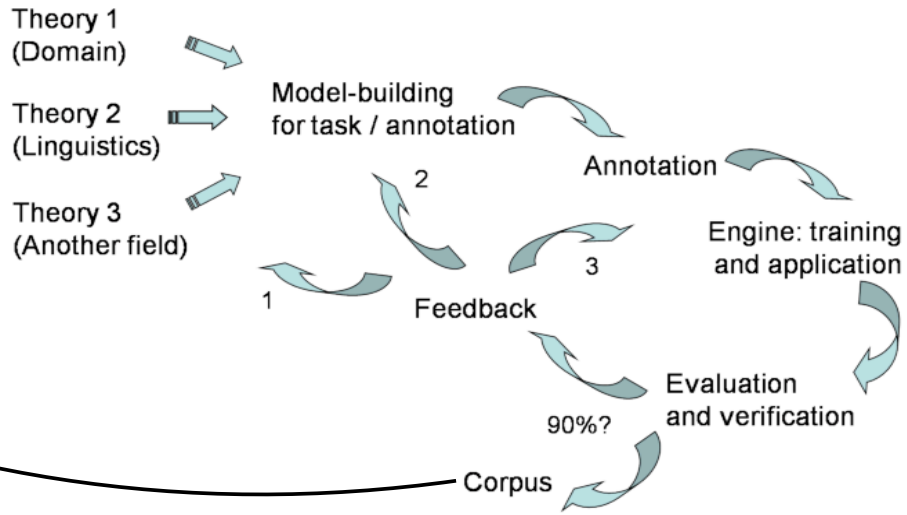
2.1.3 Idealised Supervised NLP Workflow

1. Define the task
2. Collect and annotate data
3. Prepare the data
4. Train the model
5. Evaluate and interpret results
6. Analyze outputs and errors
7. (Deploy?)



2.1.3 ~~Idealised~~ Supervised NLP Workflow

1. Define the task
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2.2 Representation + standard ML



2.2.1 Overview

> **Approach:** *Use static representations*
>
> Use *precomputed text embeddings* as numerical representations of texts, then train a *standard ML model* on top.

1. **Embed** text

- BoW, TF-IDF, Word2Vec, Sentence-BERT, …

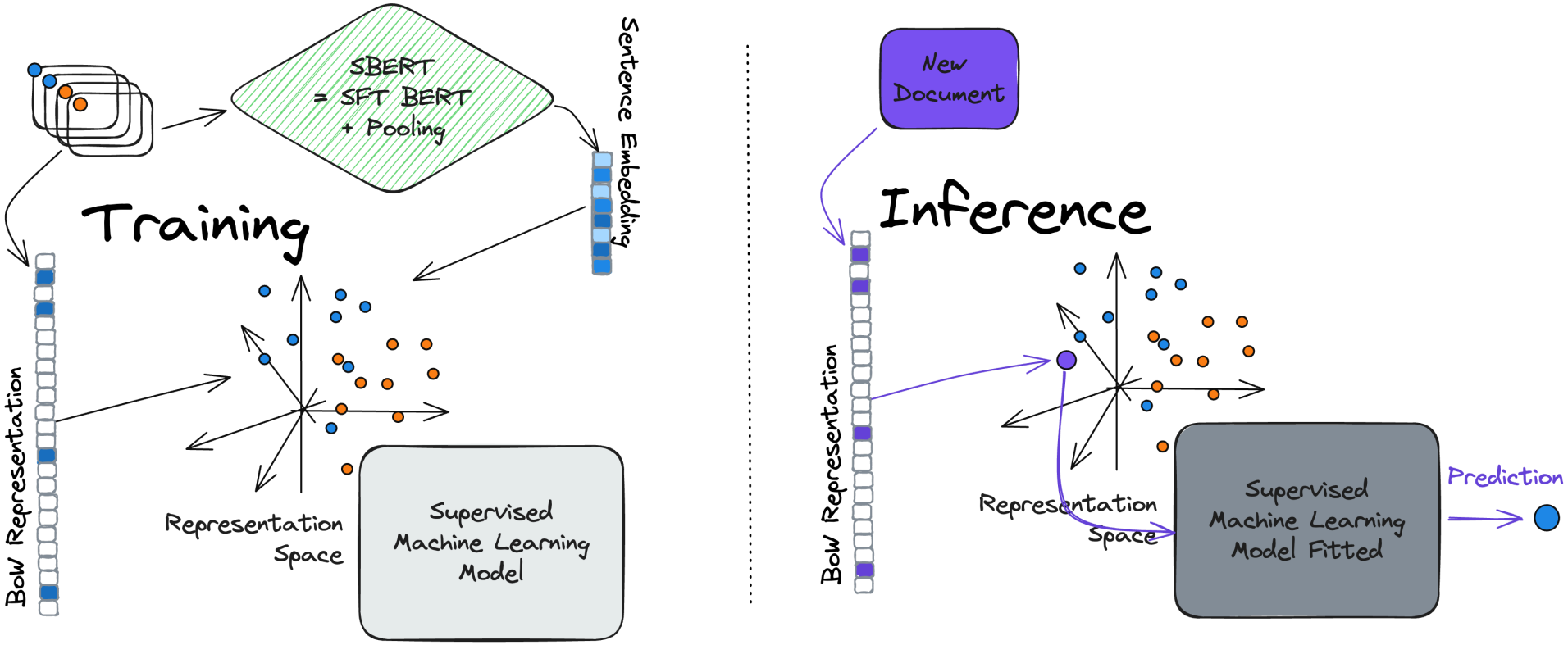
2. **Train** a standard ML model on the labelled embeddings

- Logistic Regression, SVM, Random Forest, MLP, …

3. **Predict** in the *embedding space*

- new document → frozen embedding model → trained ML model → prediction

2.2.2 Rep+ML: schematic pipeline



2.3 Supervised Fine-tuning

2.3.1 Overview

> **Paradigm Shift:** *From static features to dynamic task adaptation*
>
> Adapt a *pretrained transformer model* end-to-end to a specific downstream task by *updating its parameters* through training on labeled data.

1. Choose a **pretrained model** & add a **task-specific head**

- pre-trained LM already knows language structure (remember LM lecture)
- add a task-specific (classification, regression, …) head on top of LM

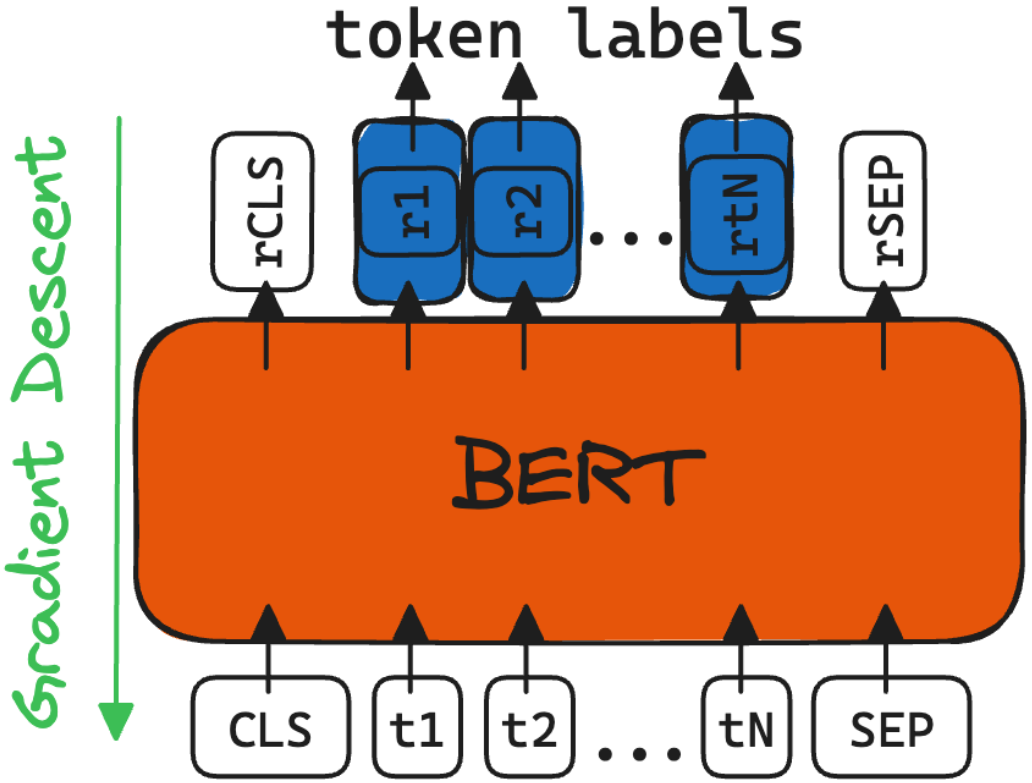
2. **Fine-tune** the model

- update the model's weights during training thanks to the labeled data

3. **Predict** directly from text

- text → fine-tuned LM → prediction

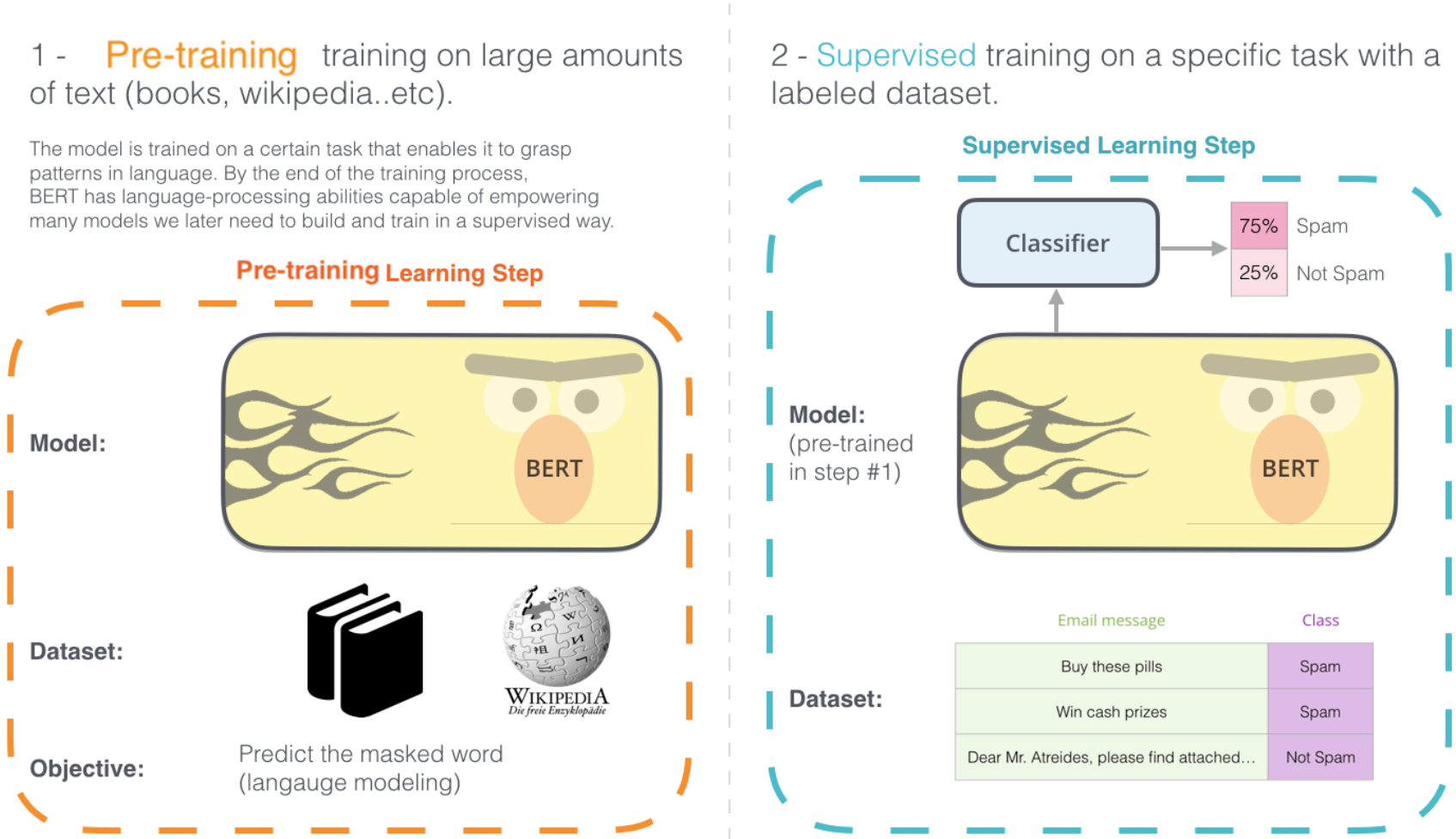
2.3.2 Standard architecture and process



Sequence labelling

- $z_t = \mathbf{W}\mathbf{h}_t + \mathbf{b}$

2.3.3 SFT Overview



Two steps of BERT development (extracted and modified from (Alammar 2018)).

2.3.4 Fine-tuning a PLM in practice

```
# Instantiate the Trainer object
trainer = Trainer(
    model=model,                         # the instantiated 🤗 Transformers
    args=training_args,                  # training arguments, defined abov
    train_dataset=train_dataset,         # training dataset
    eval_dataset=test_dataset,           # evaluation dataset (usually a va
    compute_metrics=compute_metrics      # our custom evaluation function
)

# Launch training!
trainer.train()
```

2.3.5 Effects of fine-tuning

> **Core Takeway**
>
> **Fine-tuning** transforms a *general-purpose* language model into a *specialized* tool.

- Data-efficiency
  - Adapting requires much fewer labeled examples than training from scratch (Howard and Ruder 2018)
- Semantic Space Reorganisation
  - shifts its internal vector space, clustering task-relevant concepts together
  - see e.g.; Zhou and Srikumar (2022)
- Distribution Narrowing
  - broad, generalized language distributions → specific domain or task style
- Loss of Generalisation
  - "Over"-narrowing the distribution, or relying on dataset-specific shortcuts ⇒ accuracy may drop on out-of-distribution text within the same task domain (McCoy, Pavlick, and Linzen 2019)
- *Catastrophic Forgetting* (McCloskey and Cohen 1989)
  - (Over)updating the parameters of the model can degrade its the general abilities
  - Debate and nuance in BERT, see e.g.; Merchant et al. (2020)

2.4 Summary



2.4.1 Takeaways

Representation + ML	Fine-Tuning
Freeze embeddings, train small model	Adapt all model weights
Fast, lightweight	Higher compute cost
Works with few labels	Needs more data
Easier to interpret & reuse	Task-specific, less interpretable
Strong for exploratory DH tasks	Best for high-performance NLP

Key point

Representation + ML: modular, interpretable, resource-light

Fine-Tuning: powerful, task-adaptive, but resource-intensive

Keep in mind

- *'Ground-truth'* might not be absolute
- Model choice reflects task and data
- Tune with intention
- Interpret critically

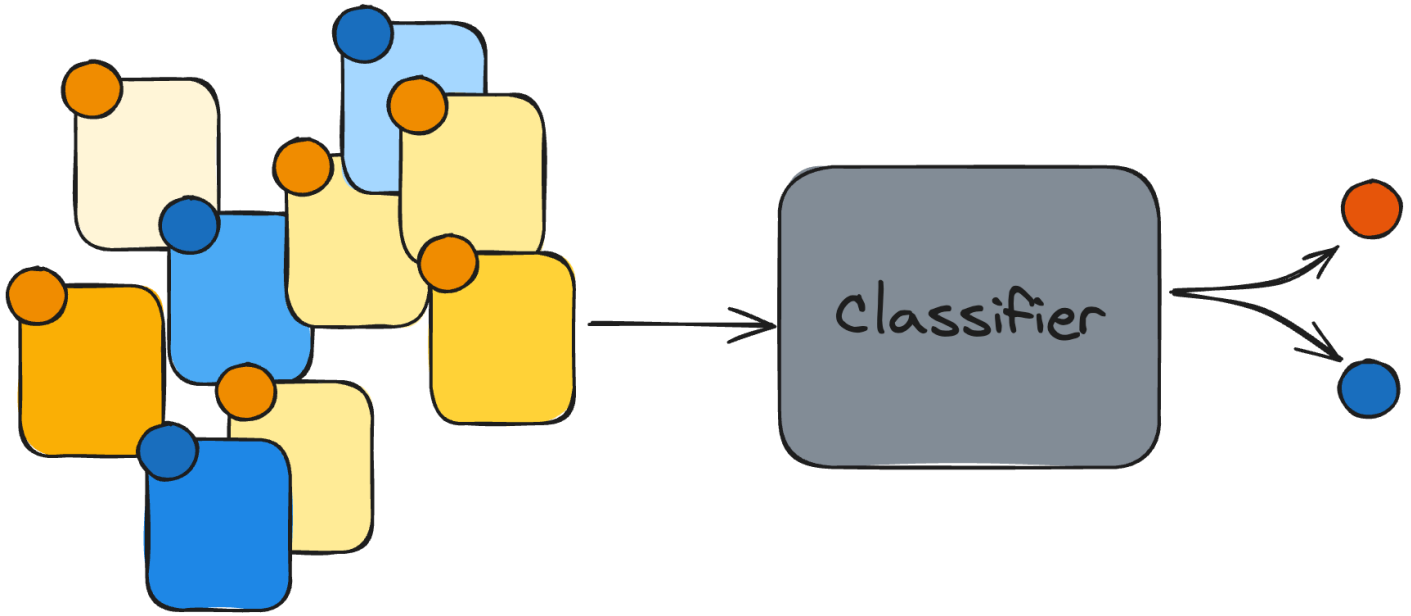
3 Classification

3.1 What is a classification task?

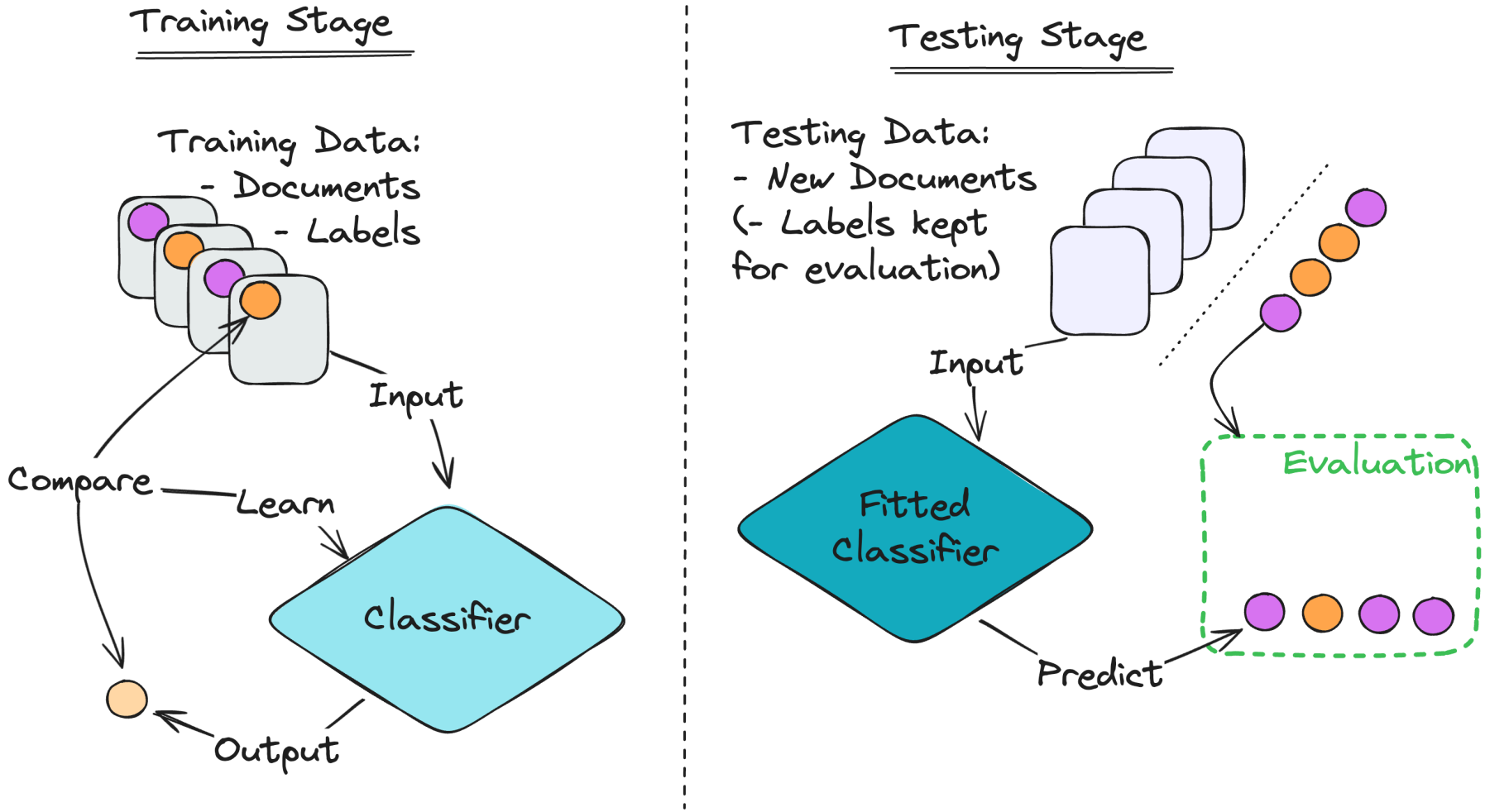
3.1.1 Definition

The Classification Task

Assign each input text to one of a set of predefined categories or classes.



3.1.2 Overview

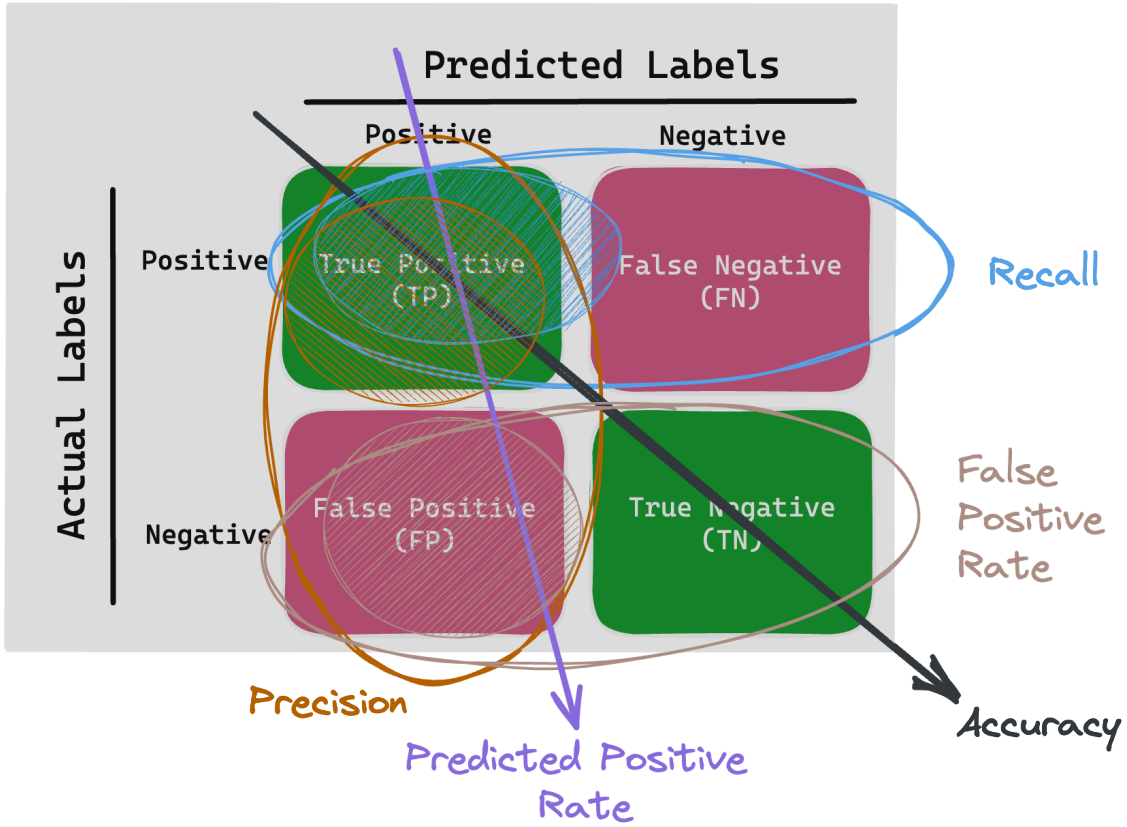


3.2 The Confusion Matrix & Performance Metrics

3.2.1 Confusion Matrix

Definition

Matrix where each cell counts examples by how the predictions compare to the ground truth.



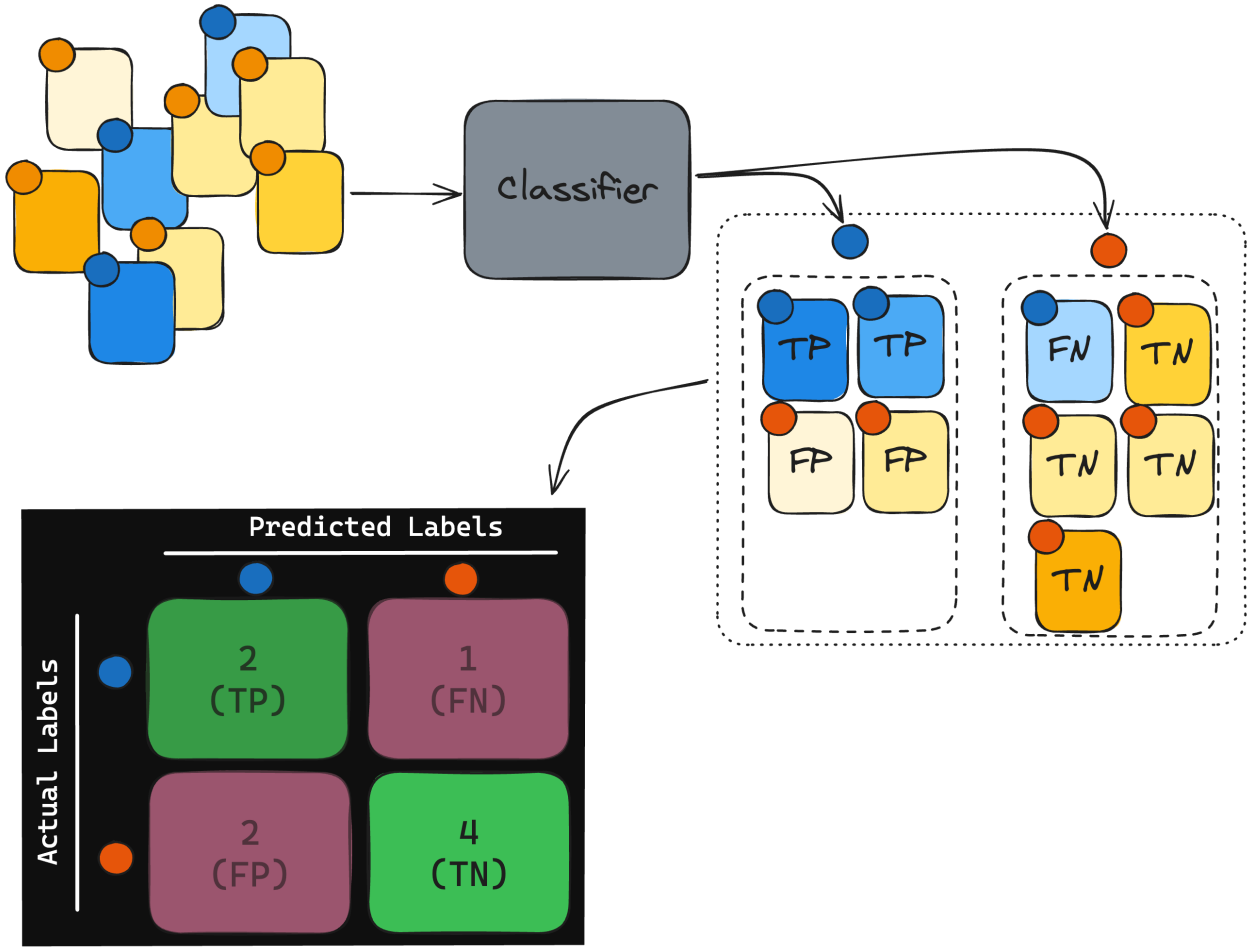
→ *Used to compute different performance metrics* (see also:)

3.2.2 Performance Metrics

Metric	Formula	Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	› Overall proportion of correct predictions
Positive Predicted Value	$\frac{TP}{TP+FP}$	› How accurate the positive predictions are (=precision)
True Positive Rate	$\frac{TP}{TP+FN}$	› Coverage of actual positive sample (=recall)
F1-score	$\frac{2\times PPV\times TPR}{PPV+TPR}$	› Harmonic mean of precision and recall
False Positive Rate	$\frac{FP}{FP+TN}$	› Proportion of incorrect positive predictions
Predicted Positive Rate	$\frac{TP+FP}{TP+TN+FP+FN}$	› Proportion of samples predicted as positive

3.2.3 Quick Example

- Accuracy
  - (2+4)/(2+1+2+4) = 6/9 ≈ 0.66
- Precision
  - 2/(2+2) = 2/4 = 0.5
- Recall
  - 2/(2+1) = 2/3 ≈ 0.66
- F1-score
  - $\frac{2\times precision\times recall}{precision+recall}$
- …

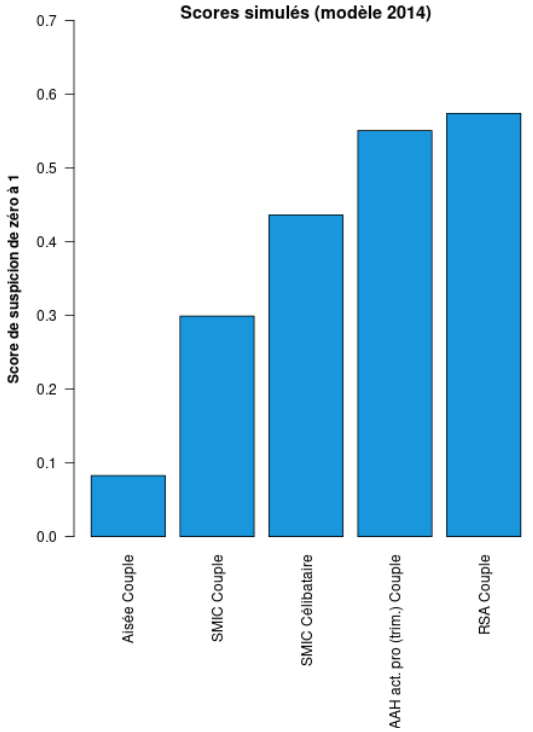


3.3 *Is that enough?*

3.3.1 Some examples…

… reinforcing structural discriminations, through:

- Reported Bias & Runaway Feedback Loops
  - Minorities targeting in predictive justice (Angwin et al. 2016); predictive policing (Lum and Isaac 2016; Rotaru et al. 2022)
- Latent Bias & Sampling Bias
  - Racial healthcare disparities (Obermeyer et al. 2019)
  - Gender bias in credit scoring (Telford 2019) and recruitment practices (Dastin 2022)
  - Classist biases in welfare systems (QdN 2023)
- …



CAF's suspicion score (extracted from (QdN 2023)).

3.4 Fairness Metrics



3.4.1 Conceptual Overview

Three core concepts (Barocas, Hardt, and Narayanan 2023):

1. **Independence** — predictions are statistically independent of sensitive attributes
   - e.g., same positive rate for all groups
2. **Separation** — error rates are similar across groups, conditional on the true label
   - e.g., same false-positive / false-negative rates for each group
3. **Sufficiency** — predictions are equally reliable across groups
   - e.g., for a given predicted probability, the actual correctness is the same

3.4.2 Underlying idea



Fairness across groups

The model should perform similarly across groups.

→ compute performance metrics for differnt groups:
let p refer to privileged group and n to non-privileged group:

$$m_p = \text{score}(TP_p, FP_p, TN_p, FN_p), \quad m_n = \text{score}(TP_n, FP_n, TN_n, FN_n)$$

→ Compute the ratio; a “fair” model should ensure (RoT: $\epsilon \approx 0.8$):

$$\epsilon \leq \frac{m_n}{m_p} \leq 1/\epsilon$$

3.4.3 Practical Measures of Fairness

Name	Metric	Concept	Interpretation (Wiśniewski and Biecek 2022)
Statistical Parity	**PPR**	Independence (equivalent)	whether both groups have equal likelihood of being *predicted as positive*
Equal Opportunity	**TPR**	Separation (relaxation)	likelihood of *correctly* recognising a *positive* is equal regardless of group
Predictive Equality	**FPR**	Separation (relaxation)	likelihood of *incorrectly* classifying a *negative* as *positive* is equal regardless of group
Predictive Parity	**PPV**	Sufficiency (relaxation)	whether *positive* are equally reliable across groups
Accuracy Difference	**ACC**	—	whether the model *performance* is *consistent* across groups

[💡 Reminder: *based on per-group performance*]

Metric	Formula	Interpretation
Predicted Positive Rate (PPR)	$\frac{TP+FP}{TP+TN+FP+FN}$	› Proportion of samples predicted as positive
True Positive Rate (TPR)	$\frac{TP}{TP+FN}$	› Coverage of actual positive sample (=recall)
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	› Proportion of incorrect positive predictions
Positive Predicted Value	$\frac{TP}{TP+FP}$	› How accurate the positive predictions are (=precision)
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	› Overall proportion of correct predictions

3.5 Key Message

- Strong Model $\not\Rightarrow$ Fair Model
- (Fair Model $\not\Rightarrow$ Strong Model)

4 Hands-on

4.1 Hands-On Challenge

Fullscreen

<Canonicity Prediction and Fairness />

The Digital Librarian v.2

📦 data_source

Excerpts of 19th/20th-centuries books from *ANRChapitres* Corpus (FR).

› 5-sentence long chunks

≈ 10 chunks X 2000 books

› canon / non-canon labels

🎯 goal

Classify the texts based on their 'canonicity' (if they are part of the literary canon or not).

→ Evaluate *performance*, but also *fairness* of the classifiers (with respect to 'author_gender').

⚙️ methods

Yours to choose!

📚 examples

“Nous sommes en route en plein midi. La chaleur est tombée. Il fait même très froid en voiture

References

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